

Unit 9, Lesson 3: Finding and Interpreting the Mean

Benchmark

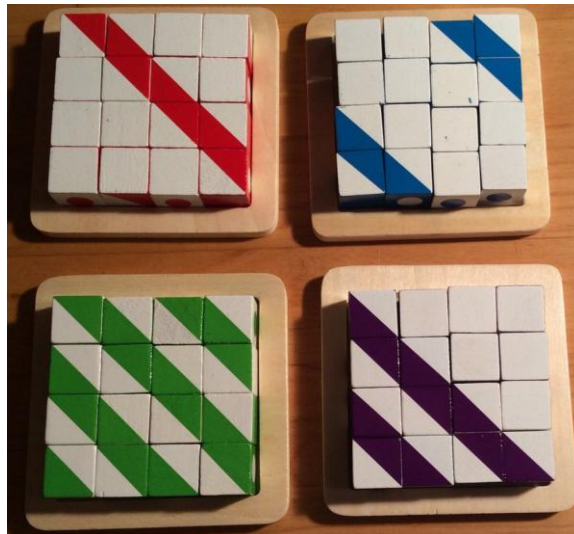
MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.

Learning Target

- ☐ I can find and interpret the mean of a data set.

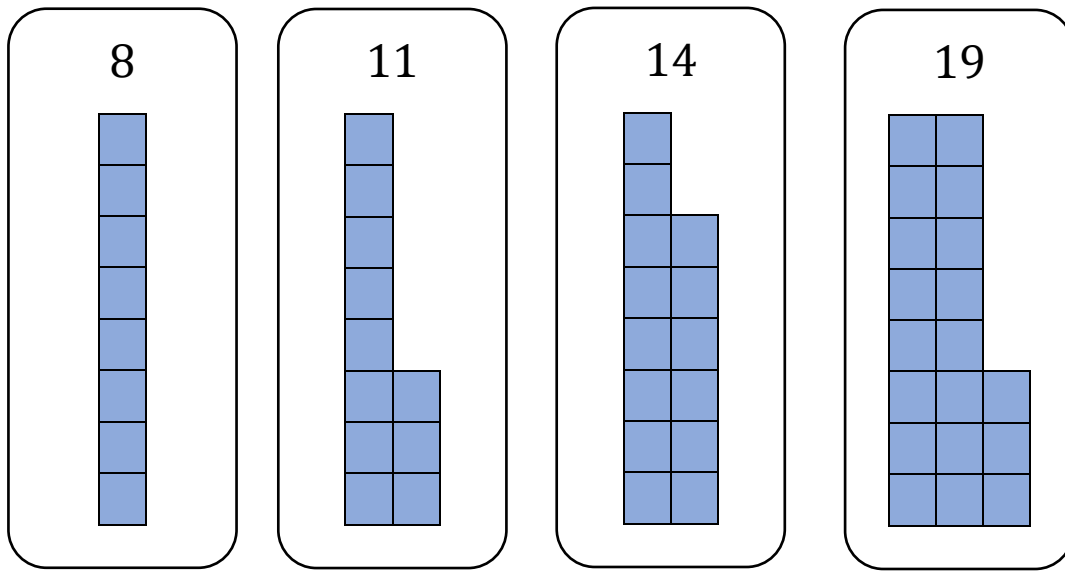
9.3.1 Warm-Up: Stripes

A picture of four sets of 16 cubes is shown.

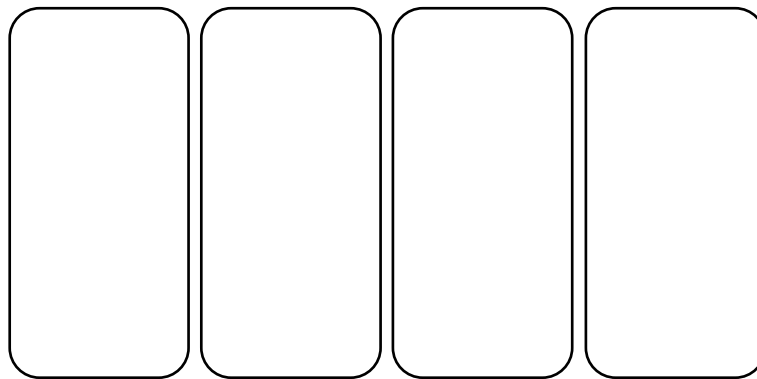


1. Which set of cubes has a set of stripes that covers the most area?
2. Explain how you determined your answer.
3. Which set of cubes has a set of stripes that covers the least area?
4. Explain how you determined your answer.

9.3.2 Exploration: Sharing Squares



1. Estimate the number of squares that will be in each group if they were distributed evenly between the four groups.
2. Use the diagram to redistribute the squares so that each of the four groups has the same number of squares.



3. Compare diagrams with your partner. Discuss how you each determined the number of squares that belong in each group. Summarize your discussion.

9.3.3 Guided Instruction: Mean of Data

By redistributing the squares in the diagram, the **mean** of the data set was found.



The **mean** is the arithmetic average of a set of numbers. It is a measure of central tendency.

Data in a numerical set is summarized using different measures that are interpreted to describe the data. One such measure used to describe quantitative data sets is the mean.

To calculate the mean, add all the values in the data set and then divide by the total number of data values.

1. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are 75, 300, 225, 400, 250, 195, 320, 45, 165, 410.
 - a. To calculate the mean time spent underwater, start by finding the _____ of the data values.
 - b. Then divide the sum by 10, because there are ten turtles.
 - c. In this sample, Red-Eared Slider turtles remain underwater for an average of _____ minutes.



Measures of center are numerical values used to describe the overall clustering of data in a set, or the overall central value of a set of data. The three most common measures of central tendency are the mean, median, and mode.

2. Data from the Pew Research Center on the percentage of adults in the U.S. who use Instagram is shown in the table.

Percentage of U.S. Adults Using Instagram	
Year	Percentage
2012	13%
2013	17%
2014	26%
2015	29%
2016	34%

- a. Describe, in words, how to find the mean percentage of adults in the U.S. who use Instagram.

Step 1:

Step 2:

- b. Find the mean percentage of adults in the U.S. who use Instagram. Think about what type of number the answer should be.

- c. Interpret the mean based on the data. Specifically address what the value of the mean states about the number of adults who use Instagram.

- d. Explain why the mean does not give information about the percentage of adults in the U.S. using Instagram today.

The **mean** of a data set is one way to describe the center of data set, by generalizing the data. In other words, it describes a typical value of the data set.

- e. Complete the statement to interpret the mean of the data set from the Pew Research Center.

Typically, about _____ of adults in the U.S. used Instagram during the years 2012 – 2016.

9.3.4 Try It: Finding and Interpreting the Mean

1. The heights of seven NBA players are listed in the table.

Player	Height (in inches)
LeBron James	81
Stephen Curry	75
Kevin Durant	82
Kobe Bryant	78
Michael Jordan	78
James Harden	77
Shaquille O'Neal	85

- a. What is the mean height, in inches, of the NBA players sampled? Round to the nearest tenth.
- b. Interpret the mean height for the seven players.
- c. Determine whether the mean should be used to make a generalization about all NBA players. Justify your answer.
2. Below is a list of the number of home runs hit in the top 10 most home-run friendly ballparks during the 2012 Major League Baseball season.

231	202
230	199
229	175
225	174
218	158

Circle the value from the data set that is closest to the mean number of home runs per ballpark. Show your work.

9.3.5 Your Turn!

1. Statistics on the total number of visitors to Florida are tracked by Visit Florida Research. The table shows the number of visitors, in millions, for each quarter for the year 2019.

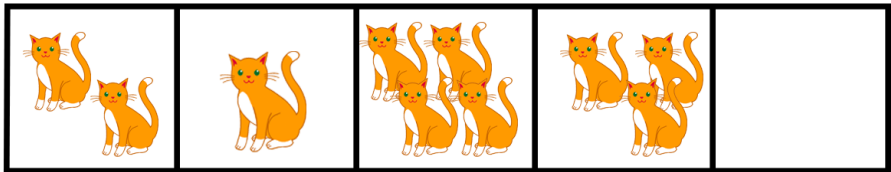
Visitors to Florida (in millions)	
Quarter	Total Visitors
Q1 (January-March)	35.79
Q2 (April-June)	32.41
Q3 (July-September)	32.40
Q4 (October-December)	30.81

- a. What is the mean number of people, in millions, who visited Florida **per quarter** in 2019? Round your answer to the hundredths place.
- b. What is the mean number of people, in millions, who visited Florida **per month** in 2019? Round your answer to the hundredths place.
- c. Interpret the mean number of visitors per quarter in terms of the data set.
2. For one week, Diego kept track of how long it took for his dad to drive him to school each morning.

Diego's Travel Time to School (in minutes)	
Day	Travel Time
Monday	22
Tuesday	18
Wednesday	11
Thursday	8
Friday	11

- a. What is the mean travel time for Diego to get to school?
- b. Interpret the mean in terms of the data set.

3. An animal shelter has 5 crates available to hold 10 kittens. The manager of the shelter wants the kittens to be distributed equally among the crates.



- a. How many kittens will end up in each crate?
- b. If the shelter receives 2 more kittens, how many kittens must be placed in each crate so they are equally distributed?
- c. The shelter wants to keep kittens equally distributed and use all five crates. Circle the number of kittens that allows the shelter to equally distribute the kittens among five crates.

12	15	6	18	20	23	9
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9.3.6 Wrap-Up: Cool Down

1. Explain what the **mean** of a data set represents.
2. What is the average height of the bars? Draw a horizontal line across the bars to show the height of the bars if you redistributed the values so that all of the bars showed the same height.

